

Preetham Dharshan Thangellapally

Software Engineer

Phone: +1 (334) 440-6012 | **Email:** preethamdharshan62@gmail.com | **Location:** Montgomery, AL | [LinkedIn](#) | [GitHub](#)

Summary

Results-driven Software Engineer with 4+ years of experience in backend development, cloud-native applications, and data engineering. Skilled in designing scalable ETL pipelines in Snowflake, building high-performance APIs and microservices using Go and Python, and deploying secure, HIPAA-compliant solutions. Proficient in AWS, Kubernetes, Docker, Terraform, and CI/CD automation, with expertise in performance optimization and ML model integration to deliver robust, data-driven systems.

Experience

Software Engineer | Snowflake, USA – Remote

Aug 2024 – Present

- Designed and deployed scalable data pipelines in Snowflake using Python (Snowpark API) and SQL, processing 5TB+ of daily transaction data with a 40% reduction in ETL job execution time.
- Engineered a real-time analytics dashboard by integrating Snowflake with AWS Lambda and React.js, enabling sub-second latency visualizations and improving decision-making speed by 60%.
- Led performance tuning for Snowflake's multi-cluster warehouses, rewriting complex SQL queries and creating materialized views, reducing AWS infrastructure query costs by \$15K/month.
- Developed a Python-based data governance tool leveraging Snowflake's dynamic data masking and row-level security, ensuring HIPAA compliance for 1M+ patient records in a healthcare client environment.
- Enhanced CI/CD pipelines using GitLab CI/CD, Docker, and Python SDK, enabling automated testing, version-controlled deployments, and zero-downtime releases while reducing deployment failures by 35%.
- Implemented infrastructure-as-code using Terraform and AWS CloudFormation, automating environment provisioning and reducing manual configuration time by 40%.
- Integrated system monitoring and logging solutions with AWS CloudWatch and ELK Stack, improving observability, proactive alerts, and reducing incident resolution time by 35%.
- Collaborated with data scientists to deploy machine learning models (scikit-learn) within Snowflake using Snowpark ML, improving fraud detection accuracy by 22% for a financial services project.

Backend Engineer (Go/Golang) | Razorpay, India

Jun 2020 – Dec 2023

- Engineered and optimized high-performance RESTful APIs using Go (Gin framework), reducing response latency by 30% and handling traffic for millions of transactions daily.
- Developed a custom JWE token layer over JWT and enforced OAuth2 standards to secure internal and external APIs, strengthening authentication and compliance with security best practices.
- Designed a concurrency-safe worker pool in Go to enhance resource utilization in the Self-KYC pipeline, increasing throughput by 25% and accelerating user verification processes.
- Integrated ML-powered OCR and coordinate prediction models for automated redaction of sensitive information in government-issued IDs, improving compliance and reducing manual intervention by 80%.
- Built a scalable file processing pipeline to standardize diverse input formats (PDF, JFIF, TIFF, BMP, GIF) into JPEG/PNG for ML model compatibility, improving processing efficiency by 40%.
- Architected and deployed microservices on AWS EKS using Docker, Kubernetes, and Helm, enabling seamless horizontal scaling and reducing deployment downtime by 50%.
- Implemented automated CI/CD pipelines using GitLab CI and managed infrastructure provisioning with Terraform, reducing environment setup time by 60% and ensuring consistent deployments.
- Designed and maintained fraud detection algorithms for dynamic user blacklisting across store-level and global scopes, improving fraud prevention measures and reducing security incidents by 35%.

Skills

Languages & Scripting: Python, Go (Golang), SQL, JavaScript (React.js), TypeScript, HTML/CSS, Shell Scripting

Data Platforms & Warehousing: Snowflake, Multi-Cluster Warehouses, Materialized Views

Backend & APIs: Gin (Go), RESTful APIs, GraphQL, OAuth 2.0, JWT, JWE Token Layer, API Documentation (Swagger/OpenAPI)

Frameworks & Libraries: Django, Gin, FastAPI, TensorFlow, Keras, OpenCV, Tkinter, Bootstrap, Snowpark API

Data Engineering & Pipelines: ETL/ELT Design, Batch & Streaming Pipelines, Change Data Capture (CDC), Data Transformation

Frameworks, Data Lineage Automation, Data Masking, Row-Level Security

Databases & Storage: PostgreSQL, MongoDB, SQLite, Redis, Amazon S3 Object Storage

Cloud & DevOps: AWS (EKS, Lambda, IAM, CloudWatch), Terraform (IaC), GitLab CI/CD, Docker, Kubernetes, Helm

Security, Compliance & Governance: HIPAA, Access Controls, Secrets Management, Audit Logging, Data Privacy Controls

Analytics, ML & Apps: scikit-learn Model Deployment, Real-Time Analytics Dashboards (React), Fraud Detection Pipelines

Performance & Cost Optimization: Query Tuning, Partitioning, Z-Ordering, Caching Strategies, Warehouse Scaling

Monitoring & Observability: Prometheus, Grafana, CloudTrail, Log Aggregation & Alerting

Collaboration & Delivery: Agile/Kanban, Git, Code Reviews, Technical Documentation, Cross-Functional Team Communication

Education

Master of Science in Computer Science

Jan 2024 – May 2025

Auburn University at Montgomery, Montgomery, AL

Bachelor of Technology in Computer Science

Jul 2018 – Aug 2022

Nalla Malla Reddy Engineering College, India

Project

Multiple Face Mask Detection | Python, OpenCV, TensorFlow, Keras, Tkinter

- Developed a real-time face mask detection system using CNN-based deep learning models, achieving 95% accuracy in classifying masked vs. unmasked faces.
- Implemented OpenCV for live video capture and face detection, with visual overlays, and designed a Tkinter-based desktop application for user-friendly interaction.
- Enhanced model generalization by applying custom image augmentation (rotation, brightness, flipping) and sourcing datasets from Kaggle for robust performance under varied conditions.